

How do Players Perceive Gender Discrimination? On the Differences of Harassment in Online Games



Abstract—Competitive environments, whether in traditional sports or competitive online gaming, often foster intense emotions and harsh language. In Competitive Online Multiplayer Games, where players are not physically present together, verbal toxicity primarily manifests through voice and text chat. As online gaming remains a predominantly male-dominated space, much of this toxicity disproportionately targets non-male players. Behind the veil of anonymity, non-male players frequently face gender discrimination, hate speech, and unwanted sexual advances. To effectively address these issues, game developers must first understand how such harassment manifests and how it is perceived by players. This work examines gender discrimination in Competitive Online Multiplayer Games through two online surveys. The first survey gathered reports from 61 non-male players who had experienced harassment, resulting in 171 coded statements describing gender discrimination. A second survey presented these statements to 281 players across all genders, who rated their perceived severity and authenticity. Our results indicate that gender discrimination is generally perceived as equally severe across all gender groups. However, notable differences are visible in how players respond to harassment, with those from marginalized groups exhibiting higher levels of rumination. This highlights the compounded impact of gender-based toxicity in online gaming. These insights provide valuable direction for game developers seeking to create more inclusive and supportive gaming environments.

Index Terms—Player Experience, Online Harassment, Gaming Communities, Survey Studies

I. INTRODUCTION

Competitive Online Multiplayer Games (COMG) are among the most popular games. While traditionally considered a male-dominated space, female players' numbers are on the rise [1]. Riot Games reported that up to 40% of its VALORANT playerbase identify as female, while the playerbase of League of Legends (LoL) is reported to be up to 30% [2]. Overall, roughly 34% of first-person shooter (FPS) gamers are reported to be female in the US in 2024 [3]. For the given genres, we can therefore assume that roughly a third of its playerbase identifies as female in 2025.

Gaming has been said to improve social interactions for players [1], [4], but at the same time, female players, next to other marginalized groups, often report experiences of harassment [5]. For female players, comments discriminating against their gender are a common occurrence in competitive online spaces. The problem persists, although approaches to minimize

toxicity in games have developed further and multiplied through the last decade. Previous work has underlined the importance of spaces free from prejudice, in which minority groups can meet and socialize [6]. After all, toxicity against minorities heavily impacts the mental health of players in that minority. If handled appropriately, COMG have the potential to offer a great range of mental health benefits to players, especially those in minority groups [6]. Notably, social support is not only an important factor for mental health after experiencing harassment, but also an existential motivating factor for players to play games [5].

Although overall gamer demographics are relatively balanced, prior research suggests that female players remain a perceived minority in Competitive Online Multiplayer Games (e.g., 34% in FPS games in 2024 [3]). Therefore, they are inherently regarded as having lower status by other players [4], [5], [7]. The growing number of female players in the traditional male-dominated space is not always being welcomed by male users, and hence sparks a problem of sexual harassment in the gaming community [8]. In fact, female players are constantly reminded of them trespassing the intended male subject position through sexual harassment [7]. Spaces dedicated to geek culture, which includes gaming communities, may exhibit the tendency to view women as either objects of sexual desire or unwelcome intruders [8].

To overcome gender discrimination in COMG, we must first understand how it is perceived. Since prior research has been conducted to observe the overall problem of harassment of female players, this work aims to examine to what degree it hinders or discourages them from playing. As such, we focus on the perceived severity of gender discrimination against female players for players of all genders. Furthermore, we aim to understand in what ways the perceived severity impacts rumination and consequent withdrawal from games. To this end, we conducted two studies with 61 and 281 participants, respectively. Participants of both studies were asked to provide insight into their perception of gender discrimination and harassment towards female players.

This paper is structured as follows: Section II reviews the literature on gender discrimination, harassment, and its impact on online gaming. Section III outlines the research questions, methodology, and data collection. Section IV presents our

findings on harassment severity and its effects. Section V addresses validity threats and discusses the implications for gaming communities and developers. Finally, Section VI concludes the study and suggests future research directions.

II. BACKGROUND AND RELATED WORK

A. Harassment in online games

The reasons behind the harassment of marginalized groups in COMG have been widely studied. Fox and Tang [7] found that social dominance orientation, the desire for power over women, and the need for heterosexual self-preservation are significant predictors of video game sexism. They explain that masculine behavior is often rewarded in gaming environments, providing male players with an opportunity to assert social dominance in ways that may not be feasible in the physical world.

The online disinhibition effect described by Suler [9] suggests that inherent characteristics, including anonymity, minimized authority or lack thereof, and a hypermasculine atmosphere, may further facilitate harassment online. Furthermore, the Social Identity Model of Deindividuation Effects (SIDE), as proposed by Postmes et al. [10], argues that the anonymity of online interactions leads users to downplay their individual identities and instead conform to salient group norms, reinforcing collective behaviors. Fox and Tang [11] add that this includes the realm of COMG, and the male players in the male-dominated space of gaming are likely to defer to their masculine identity. Suler [9] furthermore explains that people tend to feel less vulnerable about acting out, when provided the opportunity to separate their online actions from their real-life character. Often, players practice moral disengagement by employing cognitive operations that allow them to justify, rationalize, and downplay problematic behavior [12].

Examples of this effect can be found in many popular COMG. Ma et al. [6] suggest that FPS games may be the most toxic environments for female players and suggest SIDE to be a main cause. Maharani et al. [13] describe how gender discrimination in the game VALORANT is a universal experience for female players. They explain that voice chat, in which teammates can communicate with each other using their microphones, is the primary avenue for toxicity in the game.

Aguerri et al. [14] analyzed toxicity in the game LoL and concluded around 70% of played games show toxic behaviors. They further emphasize that the definition of toxicity in gaming is highly subjective, leading to differing opinions among players on what constitutes toxic behavior. Consequently, we conclude that the variation in perceived severity of gender discrimination in COMG presents an intriguing area for analysis.

The impact of harassment varies among players, leading them to adopt different coping mechanisms. According to Adinolf and Türkay [15], ignoring or muting other players is the most common strategy, which inadvertently allows toxic behavior to persist unchecked. Drenten et al. [16] further argue that this leads to a feedback loop, in which marginalized groups use coping strategies that in turn solidify the perceived dominance of the majority group. McLean [5]

suggests that the lack of societal support and the prevalence of harassment often lead female gamers to play alone, remain anonymous, or frequently change groups. The author notes that female players commonly report feelings of anxiety and loneliness. This may further contribute to the isolation of marginalized players, reducing their visibility in gaming spaces and reinforcing the perception that competitive gaming remains a male-dominated environment. As a result, toxic behaviors may go unchallenged, while the absence of diverse voices in gaming communities limits efforts to foster inclusivity. Given that emotional responses to harassment in gaming have been extensively documented in previous research, our study focuses on the players' own perception of harassment, its severity, and its relation to their self-concept.

B. Effects of Harassment in Online Games

Adinolf and Türkay [15] explain that player moderation tools, like reporting a player, are not often used when warranted, as players do not consider the tools to be transparent enough and do not believe that it will decrease other players' toxicity. The authors explain how most reported toxicity originates from people's need to attribute failure to others, and describe that the normalization of sexual harassment could create a cycle of harm for both the community and the victims.

Harassment in online gaming, much like in the real world, can lead players to ruminate on their experiences long after they occur. Fox et al. [11] explain that the frequency of sexual harassment in online video games contributes to this rumination, which involves repetitive thinking and dwelling on negative emotions following a distressing event. The authors further state that rumination over harassment can lead to withdrawal from the game and, in some cases, from gaming altogether. While harassment affects players of all genders, female players disproportionately experience more severe forms of harassment. According to Fox et al. [11], gaming is often considered a male-dominated space, leading players to downplay their individual identity in favor of defending the prevailing social identity (masculine, competitive, and aggressive). This dynamic results in the harassment of those perceived as outsiders, particularly women, aligning with the SIDE [10] model. Additionally, the authors emphasize that women respond to general harassment differently than they do to sexual harassment. They suggest that an important distinction exists between being harassed as part of a majority group and being targeted as a member of a minority or otherwise stigmatized group. Cote [17] identifies five coping strategies used by female players, with leaving online gaming being the most common.

Since the study by Fox et al. [11], the number of female gamers has continued to rise. According to The Entertainment Software Association (ESA) [18], women accounted for 37% of gamers in the USA in 2017, a figure that increased to 46% by 2024. Given this shift, we aim to explore how these effects apply to players today and whether they persist despite the increasing number of female gamers.

III. METHODOLOGY

A. Research Goal and Questions

The goal of this research is to assess how players of different genders perceive gender discrimination of female players in online gaming. Furthermore, we aim to understand how the perceived severity of gender discrimination impacts players' user experience.

To achieve our research goal using appropriate statistical measures, we employ the Goal-Question-Metric (GQM) method in line with Wohlin et al. [19] and pose the following research questions:

- RQ1** How do players perceive gender discrimination against female players in Competitive Online Multiplayer Games, and how do these perceptions differ across gender groups?
- RQ2** How does the perceived severity of gender discrimination among players relate to rumination, and how does this relationship vary across gender groups?
- RQ3** What is the impact of gender discrimination-related rumination on player experience in Competitive Online Multiplayer Games, and how does rumination relate to discouragement from playing Competitive Online Multiplayer Games and consequent withdrawal across gender groups?

To obtain answers to these questions, we defined five sets of metrics, which we assessed through online questionnaires:

- M1) Difference in severity rating for eight types of gender discrimination and real examples between gender groups
- M2) Difference in guessed realness of actual examples of gender discrimination between gender groups
- M3) Comparison of perceived severity of gender discrimination and rumination between genders
- M4) Comparison of rumination and guessed realness of actual examples of gender discrimination between genders
- M5) Comparison of rumination between genders regarding usage of player moderation tools, discouragement to play COMG, participation in game communities, and gamer identity

B. Survey Design

We conducted two surveys, one that collected examples of gender discrimination from mostly female players and one that evaluated the obtained examples by players of all genders. The questionnaires are available in our supplementary material [20].

1) *Common Methods in both Surveys:* Both surveys included a demographic questionnaire. Participants were asked which COMG they regularly played and how many hours per week they played. They were asked to indicate whether, in addition to playing, they actively participated in gaming communities, only engaged passively, or were not involved at all. We also asked them whether they fully identified as a “gamer”, only identified as one in specific contexts, or did not identify as

a gamer at all. Moreover, they were asked to state whether they have ever left a gaming group or community as a result of gender discrimination. In the second part of each survey, participants were asked to rate the severity of different types of gender discrimination on a 7-point scale.

This severity rating concerned both the impact said categories had on them personally and the degree to which they believed it violates Terms of Service (ToS). Using 5-point Likert scales, participants were asked if they agree or disagree with statements regarding their sense of belonging in the overall gaming community, as well as the state of gender discrimination within the community. Additionally, they were asked which player moderation tools against toxicity they use and how often (e.g., reporting or muting a player). We also asked them to state their agreement (5-point Likert scale) with different examples of behavior after hearing/reading sexist statements, such as defending the victim or ignoring the statement. Lastly, all participants were asked to state their gender identity, age range, and the location of the servers they usually play on.

2) *Survey 1: Methodology and Participant Selection:* For our first survey (S1), only female players were asked to answer an online questionnaire. Participants were recruited through university classes, personal networks of the authors, female-only discord communities, and collegiate esports communities. To fully understand the experiences of participants, we asked them to state quotes including gender discrimination directed at them. Participants were asked to rate the severity of the impact of those quotes on themselves or in terms of their violation of the game's Terms of Service (ToS).

To examine how players engage with the broader community, participants were asked to indicate whether and how they modify their behavior while playing in response to gender discrimination (e.g., changing their username or altering their voice).

3) *Survey 2: Methodology and Participant Selection:* For the second survey (S2), players of all genders were asked to participate. To this end, the survey was posted in gaming community discords, university classes, personal networks of the authors, and collegiate esports communities. Further, it was posted in gaming subreddits of popular COMG that agreed to the survey being posted (r/VALORANT and r/cs2).¹

The quotes of players that were provided by participants of S1 were utilized in S2. As over 170 quotes were submitted during S1, it was not possible to present all of them in S2. To select examples for S2, we randomly chose two quotes for each of the seven severity ratings, as determined by the participants in S1, resulting in a total of 11 quotes (see Table III). After the conduction of S1, quotes were categorized. For each of the three most prevalent categories, we made sure that at least one more quote was added. All participants of S2 were asked to

¹To meet requirements of the subreddit r/VALORANT, a survey version was adapted to focus solely on VALORANT, excluding other COMG and non-existent player moderation tools. Since the broader study included VALORANT players, their responses were merged. Additional questions on VALORANT-related behavior were included but are only reported in the supplementary materials as they do not directly relate to our metrics.

TABLE I: Age, Primary Region and Gender Identity of Participants

Category	S1(%)	S2(%)	Category	S1(%)	S2(%)
Age and Primary Region			Gender Identity		
Age (18–25)	70	54	Female	91	28
European Server	94	69	Male	–	61
NA Server	6	25	Agender	4	1.5
Other Regions	<1	10	Genderfluid	2	1

rate the severity of the combined 14 quotes. Furthermore, we asked them whether each quote was real or if it was fabricated.

C. Coding Procedure

a) *Analysis of Participants' Statements:* All collected examples of gender discrimination were coded throughout two coding cycles, using *in vivo* coding for the first cycle and pattern coding for the second cycle. All steps of the coding procedure were conducted in accordance with the guidelines of Saldaña [21]. The preliminary *in vivo* coding was conducted by one researcher. This involved removing all survey answers that did not contain any forms of gender discrimination and shortening the remaining answers to the relevant content. The second coding cycle was conducted by three researchers. First, one researcher checked the results from the first coding cycle and developed a pattern coding scheme. This scheme was then reviewed and discussed with a second researcher. After revising the codes, the final pattern coding scheme was used by three researchers to sort the statements into categories of gender discrimination. Notably, our coding procedure followed a multi-label approach, as each statement could contain multiple categories of gender discrimination.

b) *Inter-rater Agreement:* To determine the inter-rater agreement for the pattern coding, we calculated the Fleiss' Kappa [22] as well as the proportional agreement between the coders. The proportional agreement describes the proportion between the statements for which the raters agreed and the total number of statements. The value for the proportional agreement is 76.2%, while the value for Fleiss' Kappa is 80.6%. In accordance with Landis and Koch [23], this indicates substantial to almost perfect agreement.

D. Data Availability Statement

To enable the verifiability and reproducibility of our findings, we provide supplementary material [20]. This material includes the survey questionnaires, the results of the surveys, and our coding procedure. To ensure the privacy of our participants, all published data has been anonymized.

IV. RESULTS

A. Demographics

Overall, 342 participants filled out the surveys (S1: 61, S2: 281). 16 submissions (S1: 6, S2: 10) were excluded due to presenting pattern responses, and thus 326 submissions were analyzed. The most common age, primary region and gender identity of participants are displayed in Table I. Regular played games and gaming habits are displayed in Table II.

TABLE II: Regularly Played Games and Gaming Habits of Participants

Category	S1(%)	S2(%)	Category	S1(%)	S2(%)
Regular Games			Gaming Habits		
LoL	56	27	>5 h/week	90	86
VALORANT	35	63	>20 h/week	29	27
Counter-Strike 2	9	11	Prefer PC	98	92
Other Games	–	61			

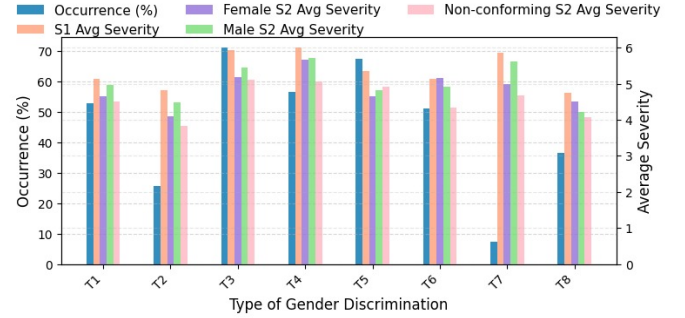


Fig. 1: Rated Average Severity and Occurrence of Types of Gender Discrimination

B. Perception of Severity of Gender Discrimination

Participants rated the severity of eight types of gender discrimination². S1 participants averaged 5.38, while S2 participants averaged 4.93. In S2, males rated higher (5.02) than females (4.87) and gender non-conforming participants (4.57). A Wilcoxon-Mann-Whitney test ($p > 0.09$) found no significant difference between gender groups in rating specific types of gender discrimination ($\alpha = 0.05$). Groups were defined by gender and type of discrimination. S1 participants reported experiencing various types of gender discrimination in the past month, with frequency weakly correlating with severity ratings ($r = 0.35$). Figure 1 shows the average severity and percentage of occurrence of types of gender discrimination by gender identity.

In S1, 42 of 61 participants submitted a total of 171 quotes from past gaming experiences. Selected quotes with participant-assigned severity ratings are shown in Table III. Quotes were categorized into eleven pattern codes, with eight quotes uncategorized, as detailed in Table IV.

Table III presents the survey quotes alongside their average perceived severity. Female participants rated quotes slightly higher (4.73) than males (4.65), while gender non-conforming participants gave the lowest average rating (4.53). A Wilcoxon-Mann-Whitney test ($\alpha = 0.05$) compared severity ratings across gender groups, defined by gender and specific quotes.

²Types are “Players Trolling or Throwing a Game because of someone’s Gender” (T1), “Players Leaving a Game because of someone’s Gender” (T2), “Verbal Harassment with Name-Calling and Insults” (T3), “Verbal Harassment with Inappropriate Sexual Messages” (T4), “Patronizing (Assuming Skill Level based on Gender) or Dismissive Comments” (T5), “Unprompted Flirting or Unwelcome Romantic Advances” (T6), “Doxxing or Targeted Harassment” (T7) and “Microaggressions” (T8).

TABLE III: Combined Severity Ratings: Participants' Ratings and Gender-Specific Ratings vs. Original Severity

ID	Quote Text	Overall	Female	Male	Other	S1
Q1	"are you female"	2.87	3.16	2.68	3.32	1
Q2	"Boosted Egirl"	4.09	4.30	3.98	4.21	2
Q3	"Do you have Instagram? Snapchat?"	3.70	3.79	3.68	3.78	3
Q4	"I didn't know we had a woman on our team. Come here! (proceeded to flirt)"	4.75	4.67	4.79	4.79	3
Q5	"slut, whore, bitch"	5.59	5.64	5.61	5.43	4
Q6	"u have big boobas"	4.24	4.23	4.34	3.79	4
Q7	"Go back to the kitchen"	4.93	4.94	4.90	5.07	4
Q8	"Hey [ingame character], are you part of the alphabet community?"	3.72	4.02	3.60	3.57	5
Q9	"Kissing sounds"	4.28	4.42	4.28	3.91	5
Q10	"I was busy thinking about getting pegged. let's make out right now"	5.23	5.05	5.36	4.89	6
Q11	"Give me your OnlyFans!"	4.81	4.83	4.88	4.43	6
Q12	"You play like a girl"	3.97	4.01	3.97	3.89	6
Q13	"your voice is makes me hard, can you suck on my dick and have e sex with me"	6.25	6.27	6.32	5.86	7
Q14	"I hope you get raped, killed and pissed on your dead body"	6.76	6.85	6.78	6.46	7

TABLE IV: Pattern Codes and their Occurrences.

Code	# Occurrences
Order to go into the kitchen or make a sandwich	43
Patronizing, "Game Over" mentality	43
Gender-specific insults	39
Sexual offers or advances	23
Messages with general sexual intent	18
Requesting to show the body, or asking for OnlyFans	10
Hate against LGBTQIA+	10
Asking for personal information	9
Sexual threats, rape threats	8
No Code	8
Derogatory sexualization	3
Gender-specific messages	1

Significant differences were found between female and male participants for Q1 and Q8.

In S2, males guessed 74% of examples were real, females guessed 73%, and gender non-conforming participants guessed 70%.

C. Rumination and Gender Discrimination

Participants were asked to state whether they ruminate after a game in which a player makes a comment that includes gender discrimination. 34% of participants agreed or strongly agreed to ruminate after such a game (S1: 70%; S2: 25%), while 47% of participants disagreed or strongly disagreed (S1: 11%; S2: 56%). In S2, male participants mostly felt neutral or disagreed, while female participants tended to feel neutral or agree to ruminate. Gender non-conforming participants generally agreed or remained neutral.

In S1, rumination and perceived severity showed a moderate pairwise Pearson correlation ($r = [0.24, 0.54]$), while in S2, the correlation was negligible to weak ($r = [0.04, 0.29]$). In S1, rumination moderately correlated with severity ratings for "Verbal Harassment with Name-Calling and Insults" ($r = 0.55$) and "Verbal Harassment with Inappropriate Sexual Messages" ($r = 0.54$). In S2, moderate correlations were found for "Verbal Harassment with Name-Calling and Insults" ($r = 0.29$) and "Microaggressions" ($r = 0.3$). The weakest associations were observed for "Players Leaving a Game Due to Someone's

Gender" (S1: $r = 0.24$; S2: $r = 0.1$) and "Doxxing or Targeted Harassment" (S1: $r = 0.27$; S2: $r = 0.04$).

Overall, participants' severity ratings of all genders showed negligible to moderate correlations with rumination (F: $r = [0.09, 0.4]$; M: $r = [0.09, 0.26]$; O: $r = [0.005, 0.534]$). A moderate correlation was observed in rumination and severity ratings of "Verbal Harassment with Name-Calling and Insults" for gender non-conforming participants ($r = 0.53$), while the weakest correlation was observed in female participants' rumination and severity rating of "Players Leaving a Game Due to Someone's Gender" ($r = 0.09$).

Participants were grouped into high-severity and low-severity groups based on their average ratings. Levene's test indicated unequal variances (S1: $W = 9.30, p = 0.0031$; S2: $W = 37.36, p < 0.0001$), so a Welch's t-test was used to assess differences in rumination between groups. The test indicated that the high-severity group ruminates significantly more than the low-severity group (S1: $t = 2.49, p = 0.0169$; S2: $t = 3.90, p = 0.0001$).

The correlation between rumination and correctly identifying real examples of gender discrimination was weak to negligible (F: $r = 0.1$; M: $r = 0.04$; O: $r = 0.2$). For Q14, the most severely rated example, rumination showed an almost moderate correlation with guessed realness in female ($r = 0.38$) and gender non-conforming ($r = 0.34$) participants, but a negligible correlation in male participants ($r = 0.024$).

In S2, rumination showed a weak correlation with severity ratings in female ($r = 0.2$) and male ($r = 0.28$) participants, but a moderate correlation in gender non-conforming participants ($r = 0.43$).

D. Effects of Rumination and Gender Discrimination

Half of S1 participants (51%) reported leaving a gaming community due to gender discrimination, compared to 23% in S2. Additionally, 59% of S1 participants agreed that gender discrimination discourages them from playing COMG, while only 28% of S2 participants felt the same.

To avoid gender discrimination, 55% of participants in S1 changed their username, 67% modified profile details (e.g., pronouns, descriptions), 16% adjusted voice settings, and 36% made no changes.

Just over half of the participants felt a sense of belonging in the gaming community (S1: 54%; S2: 52%). An ordinal logistic regression found gender was not a statistically significant predictor of belonging (S1: $r = 1.034, p = 0.087$; S2: $r = 0.2581, p = 0.207$). While S1 showed some evidence of an effect, the null hypothesis could not be confidently rejected.

Most participants felt the gaming community was not welcoming to women (S1: 68%; S2: 57%). In S1, all participants experienced at least one of the given types of gender discrimination in the past month, with 42% reporting at least four out of eight types.

In S1, 76% of participants felt gender discrimination affected their use of voice chat (29% significantly, 47% completely), while 14% reported no impact. A moderate negative correlation existed between sense of belonging and altering identity settings to conceal gender ($r = -0.31$). Chi-square analysis showed a significant link between sense of belonging and changing usernames to hide gender ($\chi^2 = 11.35, p = 0.010$), but not with other methods like changing profile information.

Pairwise correlation analysis revealed a weak to moderate association between rumination and player moderation tool usage (S1: $r = 0.24$; S2: $r = 0.27$). In S2, the correlation was moderate among female participants ($r = 0.36$) but weaker among male ($r = 0.26$) and gender non-conforming ($r = 0.23$) participants. Moderate correlations were observed for muting all players in voice (F: $r = 0.48$; M: $r = 0.35$; O: $r = 0.35$) and text chat (F: $r = 0.47$; M: $r = 0.16$; O: $r = 0.16$). Overall, female participants exhibited stronger correlations between rumination and moderation tool usage, particularly for muting and reporting features (mean $r = 0.36$), compared to male ($r = 0.26$) and gender non-conforming ($r = 0.23$) participants. A moderate correlation was found between rumination and the feeling of discouragement from playing COMG among participants (S1: $r = 0.47$; S2: $r = 0.56$). In S2, The correlation was strongest among gender non-conforming participants ($r = 0.76$), followed by female participants ($r = 0.62$), while male participants showed a notably weaker correlation ($r = 0.30$).

An ordered Logistic Regression found no strong evidence that participation level in communities outside the game influences rumination (S1: $r = 0.03, p = 0.916$; S2: $r = 0.11, p = 0.478$), including when analyzed by gender in S2 (F: $r = 0.03, p = 0.93$; M: $r = 0.195, p = 0.33$; O: $r = -0.29, p = 0.48$). After conducting a Welch's t-test, no significant relationship was found between identifying as a "gamer" and rumination.

V. DISCUSSION

A. RQ1: Perception of Gender Discrimination

The gender distribution in our study aligns with previous statistics on female representation in COMG, with approximately a third of participants identifying as female [3]. While previous research on harassment and rumination was conducted when the number of female players was lower, our study reflects the 2025 player distribution, offering a more up-to-date perspective. In 2017, Fox et al. [11] found that players of all

genders acknowledged the harassment of female players as a problem. Our findings indicate that this perception remains unchanged. Male participants, as the majority group, rated the severity of gender discrimination similarly to female and gender non-conforming participants. This applied to both the perceived severity of different types of gender discrimination and the actual examples provided by participants in S1. The slightly higher ratings of actual examples by female participants suggest that, since they encounter similar experiences more frequently, concrete examples feel more severe compared to abstract category labels. Fox et al. [11] suggest that male and female players, as a marginalized group, perceive harassment differently. Our findings indicate that while the frequency of harassment experienced by players has only a weak influence on their overall perceived severity of harassment, it does have an impact on how they perceive the severity of specific types of harassment.

Despite shifts in gender distribution, the premise of similarly perceived severity across genders remains consistent. As noted by Aguerri et al. [14], the definition of toxicity and harassment remains highly subjective among players. However, players of all genders appear to perceive the severity of gendered discrimination similarly.

Finding 1: Players across all gender groups recognize gender discrimination against female players as a persistent issue. Despite shifts in gender distribution, the perceived severity of harassment remains consistent across genders, with only slight differences based on exposure.

B. RQ2: Perceived Severity and Rumination

Fox et al. [11] proposed that individuals ruminate on experiences of harassment differently based on their identified gender. Our findings indicate that this pattern persists, despite changes in the gender distribution of players. Overall, participants demonstrated a tendency not to ruminate after experiencing or witnessing gender discrimination. However, female and gender non-conforming participants exhibited higher levels of rumination compared to male participants, who represent the perceived majority in COMG. This finding aligns with previous research indicating that marginalized groups respond to such experiences differently than the dominant group [5], [11], [14].

Our analysis indicates that players who perceived gender discrimination as more severe exhibited higher levels of rumination, with this effect being stronger in S1 than in S2. Victims of harassment were particularly prone to ruminate when experiencing severe discrimination. Notably, "Verbal Harassment with Name-Calling and Insults" and "Verbal Harassment with Inappropriate Sexual Messages" were rated as the most severe and showed some of the strongest correlations with rumination. Verbal harassment remains one of the most prevalent forms of gender discrimination in COMG, particularly affecting players from marginalized groups [6], [7], [11]. This finding aligns with our hypothesis that more frequent forms of harassment are perceived as more severe. Our findings support

Aguerri et al. [14], highlighting the subjective nature of toxicity in gaming. The findings furthermore suggest underlying factors that shape how players perceive and respond to different forms of harassment.

The perceived severity of gender discrimination and whether players have been victimized appears to have a greater impact on their tendency to ruminate than their gender.

Finding 2: Rumination is more influenced by the perceived severity of discrimination and victimization than by gender. However, female and gender non-conforming participants tend to ruminate more, especially in response to severe verbal harassment.

C. RQ3: Impact of Gender Discrimination and Rumination

Our findings support McLean [5], showing that harassment victims, especially female players, are more likely to leave gaming groups and feel discouraged from playing. A common strategy to avoid harassment is modifying usernames and profile information to conceal gender. Gender concealment through username changes is particularly linked to players' sense of belonging in the gaming community. However, unlike previous research suggesting that female players are perceived as outsiders – both by others and themselves – our findings show that female players, like participants from other groups, similarly self-identify as gamers.

Our findings suggest that players who ruminate on harassment are more likely to use moderation tools, particularly muting other players in text or voice chat. Overall, rumination increases discouragement from gaming due to gender discrimination, aligning with Fox et al. [11], who found it contributes to withdrawal. This effect persists despite shifts in gender distribution, and is most pronounced among gender non-conforming players. While gender does not directly influence perceived harassment severity, marginalized players who experience harassment tend to ruminate more, increasing their likelihood of withdrawing from gaming.

The withdrawal from gaming and identity concealment to avoid harassment may reinforce majority dominance, creating a cycle where marginalized groups hide while the majority remains unaware of their presence. This pattern aligns with the hypothesis of maladaptive boundary crossing proposed by Drenten et al. [16], wherein marginalized individuals navigate online spaces in ways that ultimately limit their visibility and inclusion.

Furthermore, this cycle presents a potential risk to the gaming community by deepening the imbalance in representation, which could reinforce and amplify perceptions of inequality. However, our findings indicate that participation levels in external gaming communities do not vary based on players' tendency to ruminate. This aligns with Ma et al. [6] and McLean [5], highlighting the significance of external gaming communities for players, suggesting that these spaces may provide a form of support and engagement that remains unaffected by in-game experiences of harassment.

Finding 3: Rumination increases discouragement from gaming, especially among marginalized groups, reinforcing the majority of the dominant group as players conceal their identities. However, engagement in external gaming communities remains stable, offering inclusive and encouraging environments.

D. Threats to Validity

As empirical research, our work is subject to several threats to validity. In the following, we discuss these limitations in accordance with Wohlin et al. [19]:

a) *Construct Validity:* As a study designed to investigate the perception of hateful behavior, our findings might be subject to social desirability bias. In particular, some participants might have rated gender discriminatory behavior as severe because they thought it was expected of them. To mitigate this, we explicitly asked participants to state their actual opinions and behavior. Furthermore, we told them that the study was not about judging their character, but rather about understanding the general behavior of the gaming community. However, we cannot claim that every participant's answers were completely unbiased in that regard.

b) *Internal Validity:* Coding procedures, such as the one conducted for this work, are typically subject to experimenter bias, as different coders may arrive at different conclusions for the same underlying data. To address this, we conducted the coding procedure with three researchers, one of which had no research context in gaming and was therefore not personally invested. Furthermore, after calculating the inter-rater agreement, we discussed the remaining disagreements until all conflicts were resolved.

c) *Conclusion Validity:* Whenever we examined data for correlations or statistically significant observations, we used appropriate statistical measures. In particular, we used Pearson's correlation coefficient, ordinal logistic regression, the chi-square test, Levene's test, Welch's t-test and the Mann-Whitney U test. All findings were evaluated at the 5% significance level. Hence, we are confident that our conclusions are reasonable regarding the data we collected.

We calculated the inter-rater agreement for our coding procedure using proportional agreement and Fleiss' Kappa. We achieved a proportional agreement of 76.2% and a Fleiss' Kappa value of 80.6%, indicating substantial to almost perfect agreement. Notably, our labeling process allowed multiple labels for each statement, meaning that a high inter-rater agreement was harder to achieve. As such, we are confident that the categories of gender discrimination identified in this work are a robust representation of the data we collected.

d) *External Validity:* The external validity of our work is limited by the diversity of our participants' demography. The first survey deliberately included only non-male participants, while the second surveyed all genders, with 61% male respondents – consistent with online gaming communities, where males are the majority. As such, we are confident that our

results are representative of larger online gaming communities regarding gender identity.

Geographically, most participants, especially in the first survey, were from Europe, with few from the US and almost none from Asia. This limits generalizability to global gaming communities, as cultural differences may significantly influence perceptions and experiences of online harassment.

Our recruitment method likely introduced selection bias, over-representing players aware of or affected by gender discrimination, while under-representing those indifferent to or engaging in it. Additionally, we lacked access to players who left gaming due to harassment, meaning our data may not capture the most severe cases of withdrawal. This imbalance could have skewed the severity ratings, highlighting the need for more representative sampling – such as randomized or stratified recruitment – in future research.

VI. CONCLUSION AND FUTURE WORK

This paper examined the perceived severity of gender discrimination and consequent withdrawal from Competitive Online Multiplayer Games. Our findings indicate that players across all gender groups perceive harassment as similarly severe, with only slight variations based on exposure, highlighting a general awareness of the issue. Players' rumination is influenced more by perceived severity than by gender, though marginalized groups tend to ruminate more, supporting prior research [11]. Severe verbal harassment plays a key role, increasing discouragement from gaming and communication, ultimately leading to withdrawal. This withdrawal reinforces majority dominance as marginalized players conceal their identities. Notably, these patterns have remained largely unchanged over the past eight years. However, engagement in external gaming communities remains stable, offering inclusive and encouraging environments.

In line with Ma et al. [6] and McLean [5], we highlight the importance of fostering inclusive and supportive gaming communities. Strengthening spaces where players can openly share, discuss, and process experiences of harassment may help mitigate rumination and its negative effects. Developers and community managers should focus on enhancing moderation tools, promoting inclusive environments, and supporting community-driven initiatives that empower marginalized players and encourage positive social interactions.

Future research should examine how harassment affects other marginalized groups and assess the effectiveness of community-driven interventions in mitigating player rumination. Longitudinal studies could provide deeper insight into how rumination evolves, offering a clearer picture of the long-term mental health impacts of gender discrimination. Additionally, such studies could explore alternative explanations, such as individual baseline tendencies for rumination. Investigating gender discrimination across diverse game genres – beyond competitive multiplayer games – may also yield valuable insights for developers seeking to design more inclusive environments.

REFERENCES

- [1] Helena Cole and Mark D Griffiths. Social interactions in massively multiplayer online role-playing gamers. *Cyberpsychology & behavior*, 10(4):575–583, 2007.
- [2] Kiko Badillo. How Many Women Are Esports Gamers?, 2024. available at: <https://playtoday.co/blog/stats/how-many-women-are-esports-gamers/> (accessed: 2025-03-11).
- [3] Statista. Distribution of first-person shooter (FPS) gamers in the United States in 2024, by gender, 2024. [<https://www.statista.com/statistics/1129301/shooter-gamers-gender/>] (accessed: 2025-03-11).
- [4] Linda K Kaye and Charlotte R Pennington. “Girls can’t play”: The effects of stereotype threat on females’ gaming performance. *Computers in Human Behavior*, 59:202–209, 2016.
- [5] Lavinia McLean and Mark D Griffiths. Female gamers’ experience of online harassment and social support in online gaming: A qualitative study. *International journal of mental health and addiction*, 17, 2019.
- [6] Lloyd Elle Ma, Camille Dickson-Deane, William Raffae, Aurora Roar Murphy, and Jaime Garcia. Gaming for Equity: The Power of Diversity within Gender and Race in Gamers. In *2024 IEEE Conference on Games (CoG)*, pages 1–8, 2024.
- [7] Jesse Fox and Wai Yen Tang. Sexism in online video games: The role of conformity to masculine norms and social dominance orientation. *Computers in human behavior*, 33:314–320, 2014.
- [8] Adrienne Massanari. #GamerGate and The Fappening: How Reddit’s algorithm, governance, and culture support toxic technocultures. *New media & society*, 19(3):329–346, 2017.
- [9] John Suler. The online disinhibition effect. *Cyberpsychology & behavior*, 7(3):321–326, 2004.
- [10] Tom Postmes, Russell Spears, and Martin Lea. Breaching or building social boundaries? SIDE-effects of computer-mediated communication. *Communication research*, 25(6):689–715, 1998.
- [11] Jesse Fox and Wai Yen Tang. Women’s experiences with general and sexual harassment in online video games: Rumination, organizational responsiveness, withdrawal, and coping strategies. *New media & society*, 19(8):1290–1307, 2017.
- [12] Lucy Sparrow, Martin Gibbs, and Michael Arnold. Apathetic villagers and the trolls who love them: Player amorality in online multiplayer games. In *Proceedings of the 31st Australian Conference on Human-Computer-Interaction*, pages 447–451, 2019.
- [13] Anissa Maharani, Virienia Puspita, Raden Army Aurora, and Nico Wiranito. Understanding Toxicity in Online Gaming: A Focus on Communication-Based Behaviours towards Female Players in Valorant. *Journal Syntax Admiration*, 5(5):1559–1567, 2024.
- [14] Jesús C Aguerri, Mario Santisteban, and Fernando Miró-Llinares. The enemy hates best? Toxicity in league of legends and its content moderation implications. *European Journal on Criminal Policy and Research*, 29(3):437–456, 2023.
- [15] Sonam Adinolf and Selen Turkey. Toxic behaviors in Esports games: player perceptions and coping strategies. In *Proceedings of the 2018 Annual Symposium on computer-human interaction in play companion extended abstracts*, pages 365–372, 2018.
- [16] Jenna Drenten, Robert L Harrison, and Nicholas J Pendarvis. More gamer, less girl: gendered boundaries, tokenism, and the cultural persistence of masculine dominance. *Journal of Consumer Research*, 50(1):2–24, 2023.
- [17] Amanda C Cote. “I can defend myself” women’s strategies for coping with harassment while gaming online. *Games and culture*, 12(2), 2017.
- [18] Entertainment Software Association. Essential Facts About the Computer and Video Game Industry, 2017. available at: <https://www.theesa.com/wp-content/uploads/2024/02/2017-EF-FINAL.pdf> (accessed: 2025-03-11).
- [19] Claes Wohlin, Per Runeson, Martin Höst, Magnus C Ohlsson, Björn Regnell, Anders Wesslén, et al. *Experimentation in software engineering*, volume 236. Springer, 2012.
- [20] [REDACTED] Supplementary Material, 2025. available at [REDACTED]
- [21] Johnny Saldaña. *The Coding Manual for Qualitative Researchers*. SAGE Publications Inc., Thousand Oaks, CA, USA, 2nd edition, 2013.
- [22] Joseph L Fleiss. Measuring nominal scale agreement among many raters. *Psychological Bulletin*, 76(5):378–382, 1971.
- [23] J. Richard Landis and Gary G. Koch. The Measurement of Observer Agreement for Categorical Data. *Biometrics*, 33(1):159–174, 1977.